

Project Title: MedAssist

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MEDASSIST:

AN AI-ASSISTED HEALTHCARE TRIAGE AND PATIENT MANAGEMENT PLATFORM

Abstract:

MedAssist is a web-based healthcare triage and patient management platform designed to organize patient information, streamline preliminary case assessment, and improve coordination between patients and healthcare professionals. The system allows patients to create profiles, enter symptoms and relevant medical information, upload medical reports, request appointments, and track their healthcare interactions through a centralized platform. Doctors can securely access assigned cases, review patient information, examine medical history and uploaded reports, manage appointments, and update case records. The platform incorporates a machine-learning-based triage module that analyzes structured patient information and categorizes cases according to predefined urgency levels such as Routine, Needs Medical Review, and Urgent Review. The ML component functions as a decision-support mechanism and does not replace professional medical judgment. The frontend is developed using React and Tailwind CSS, while Node.js and Express.js provide REST-based backend services. PostgreSQL is used for structured relational data storage, and Python with FastAPI is used to provide the machine-learning inference service. JWT-based authentication and role-based authorization provide controlled access for patients, doctors, and administrators. The system can also incorporate secure cloud storage for medical documents and an explainability layer using SHAP to provide insight into important factors contributing to model predictions. MedAssist aims to provide an organized, secure, scalable, and technology-driven approach to healthcare case management and AI-assisted triage.

Key Features:

1. Role-Based Registration and Login:

Supports separate account access for Patient, Doctor, and Administrator roles with JWT-secured authentication and role-based authorization.

2. Patient Profile and Medical History:

Patients can maintain personal information, relevant medical history, previous assessments, and healthcare-related records through a centralized profile.

3. Symptom-Based Health Assessment:

Patients can submit symptoms, symptom duration, severity, and other relevant information through a structured assessment interface.

4. ML-Based Triage Classification:

The system analyzes structured assessment information using a machine-learning model and assigns an urgency category such as Routine, Needs Medical Review, or Urgent Review.

5. Explainable AI Support:

The system can provide feature-level explanations for ML predictions using explainability techniques such as SHAP, helping users and administrators understand important factors contributing to model outputs.

6. Doctor Dashboard:

Authorized doctors can view assigned cases, patient information, medical history, assessment results, medical reports, and ML-generated triage information.

7. Case Review and Status Management:

Doctors can review cases, add professional notes, update case status, and record their final assessment independently of the ML-generated result.

8. Appointment Management:

Patients can request appointments, view available slots, and monitor appointment status, while doctors can manage schedules and appointment requests.

9. Medical Report Management:

Patients can upload medical documents and reports through secure cloud storage, while access to uploaded files is controlled according to user roles.

10. Secure Backend APIs:

Node.js and Express.js provide REST APIs with authentication middleware, validation, authorization, error handling, and protected access to sensitive operations.

11. Relational Database Management:

PostgreSQL stores structured information including users, patients, doctors, symptoms, assessments, appointments, reports, predictions, reviews, and audit records.

12. Administrative Monitoring and Analytics:

Administrators can monitor registered users, assessments, appointments, case categories, system activity, and overall platform statistics through a centralized dashboard.

13. Audit and Activity Tracking:

Important actions such as case updates, record modifications, and access-related events can be maintained through audit logs for improved accountability and traceability.

14. Extensible Architecture:

The platform can be extended with healthcare interoperability standards, notifications, teleconsultation, additional ML models, mobile applications, and integration with external healthcare systems.

Literature Survey (Existing System / Proposed System)

Existing System:

In many conventional healthcare workflows, patient information is distributed across paper records, messaging applications, individual hospital systems, appointment platforms, and manually maintained databases. Patients may provide symptoms through informal communication channels, while healthcare professionals may need to collect information from multiple sources before reviewing a case. This fragmentation can make healthcare information difficult to organize and can reduce visibility into previous assessments and appointments. Existing digital healthcare applications may provide appointment booking or electronic records but may not integrate structured symptom assessment, machine-learning-based triage, explainability, case management, and role-based workflows within one platform. AI-based healthcare systems also require careful handling because inaccurate, biased, or poorly explained outputs can create risks when used without appropriate human oversight. The World Health Organization emphasizes human autonomy, safety, transparency, explainability, accountability, privacy, and equity in the development and use of AI for health.

Proposed System:

The proposed MedAssist system introduces a centralized digital platform that integrates patient management, structured symptom assessment, appointment coordination, medical document management, and machine-learning-assisted triage. Patients can enter relevant health information through a structured interface, while doctors can securely review cases and maintain records through role-specific dashboards. A machine-learning service analyzes selected assessment features and generates an urgency classification that acts as decision support rather than an autonomous medical diagnosis. The system maintains the ML prediction separately from the doctor's professional review, allowing human oversight and comparison between automated assistance and professional assessment. PostgreSQL provides structured relational storage, while Node.js and Express.js manage the primary backend APIs and Python/FastAPI provides the ML inference service. The architecture can also be designed with healthcare interoperability in mind using standards such as HL7 FHIR for future exchange of structured healthcare information.

Research Gap:

Existing healthcare applications commonly address individual functions such as appointment booking, electronic patient records, or symptom-based information systems. However, there is scope for an integrated academic platform combining structured patient management, relational database design, secure role-based workflows, ML-assisted triage, explainable

predictions, appointment management, and centralized case tracking. Another important gap is maintaining human oversight when using machine-learning outputs in healthcare workflows. MedAssist addresses this gap by separating the automated triage result from professional review and maintaining an auditable record of both. The proposed system therefore combines software engineering, database management, machine learning, explainable AI, security, and healthcare workflow design within a single platform.

Objectives:

1. To develop a centralized web-based platform for structured healthcare case and patient management.
2. To implement secure authentication and role-based authorization for patients, doctors, and administrators.
3. To design and implement a normalized relational database using PostgreSQL.
4. To develop REST APIs for patient profiles, assessments, appointments, medical reports, case management, and administrative operations.
5. To develop a machine-learning model for categorizing healthcare cases according to predefined urgency levels.
6. To compare multiple classification algorithms using appropriate evaluation metrics such as precision, recall, F1-score, accuracy, and confusion matrix.

Tools / Hardware & Software Requirements

Front-End Development

1. HTML5
2. CSS3
3. JavaScript
4. React.js
5. Tailwind CSS

Back-End Development

1. Node.js
2. Express.js
3. REST APIs
4. Python

5. FastAPI

Machine Learning

1. Python
2. NumPy
3. Pandas
4. Scikit-learn
5. SHAP

Database

1. PostgreSQL

Authentication and Security

1. JSON Web Token (JWT)
2. Password Hashing
3. Role-Based Access Control
4. Input Validation
5. Audit Logging

Storage

1. Cloudinary / Firebase Storage / S3-Compatible Storage

Development and Testing

1. Git
2. GitHub
3. Postman
4. Docker
5. Visual Studio Code

Hardware Requirements

1. Processor: Intel i3 / AMD equivalent or above
2. RAM: Minimum 8 GB recommended
3. Storage: Minimum 20 GB free space
4. Internet Connection: Required for APIs, cloud services, deployment, and storage
5. Device: Laptop/Desktop running Windows, Linux, or macOS

Machine Learning Methodology

The ML module will use a structured healthcare assessment dataset containing relevant patient and symptom attributes and corresponding triage categories. Data preprocessing will include handling missing values, encoding categorical variables, feature selection, normalization or scaling where required, and appropriate training and testing dataset division.

Multiple supervised classification algorithms may be evaluated, including Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting methods. The models will be compared using accuracy, precision, recall, F1-score, and confusion matrix. ROC-AUC may also be considered where appropriate.

The selected model will be exposed through a FastAPI service. When a patient completes an assessment, the application backend can send the relevant structured features to the ML service. The returned triage category will be stored together with the model version and timestamp in PostgreSQL.

The system can use SHAP or a comparable explainability method to identify influential features associated with an individual model output. The final system will clearly distinguish ML-generated triage assistance from professional medical review.

No performance value will be assumed before experimentation. Final model performance will be reported using the results obtained during training and testing.

Security and Privacy

Since MedAssist handles sensitive healthcare-related information, security will be incorporated throughout the application architecture. JWT-based authentication will be used for secure access to protected APIs, while role-based authorization will restrict access according to the user's role.

Passwords will be securely hashed before storage. Input validation and controlled API access will be implemented to reduce common application security risks. Medical reports will be stored using controlled cloud-storage access, while database records will maintain appropriate ownership and access relationships.

Audit logs can be maintained for important actions such as record updates, case reviews, and administrative operations. The system will be developed as an academic prototype and will not claim regulatory or clinical compliance without formal assessment against the applicable requirements.

Base Paper Details

The project is based on research literature related to artificial intelligence in healthcare, clinical decision-support systems, machine-learning-based classification, digital healthcare management, and responsible AI.

The World Health Organization's publication "Ethics and Governance of Artificial Intelligence for Health" discusses the opportunities and risks associated with AI in healthcare and establishes principles related to human autonomy, safety, transparency, explainability, accountability, inclusiveness, and sustainability.

This research provides a foundation for designing MedAssist as a decision-support platform rather than an autonomous medical decision-maker. The project applies these concepts to a practical web-based healthcare management system by combining structured patient information, machine-learning-based triage, explainability, secure access, and professional review.

Reference:

World Health Organization. (2021). *Ethics and Governance of Artificial Intelligence for Health.*

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Other Reference Research Insights

- Digital healthcare systems benefit from structured and centralized management of patient information.
- Machine learning can support classification and decision-support tasks when appropriate datasets, preprocessing, validation, and evaluation procedures are used.
- Healthcare AI requires consideration of privacy, transparency, reliability, fairness, accountability, and human oversight.
- Explainable AI techniques can provide additional information about the factors influencing machine-learning predictions.
- Role-based access control is important when different stakeholders require different levels of access to healthcare information.
- Relational databases provide structured management of interconnected patient, assessment, appointment, and healthcare records.
- HL7 FHIR provides a standardized framework for exchanging healthcare information between systems.

Expected Outcomes

The proposed project is expected to produce a functional web-based healthcare management platform with dedicated patient, doctor, and administrator interfaces. The system will provide secure authentication, structured patient records, symptom assessment, appointment management, medical document handling, and ML-assisted triage.

The project will also provide experimental evaluation of multiple machine-learning models and integrate the selected model into the application through an API-based architecture. The final system will demonstrate the integration of Full Stack Development, SQL Database Management, Machine Learning, Explainable AI, REST APIs, Authentication, Security, and System Design within a single major project.

Future Scope

The proposed system can be extended with:

1. HL7 FHIR-based healthcare interoperability.
2. Mobile application support.
3. Telemedicine and video-consultation integration.
4. Real-time notifications.
5. Integration with authorized wearable or IoT health data.
6. Multilingual patient interfaces.
7. Additional validated machine-learning models for specific healthcare workflows.
8. Advanced healthcare analytics and administrative dashboards.
9. Model monitoring and data-drift detection.
10. Integration with institutional healthcare systems through standardized APIs.
11. Secure integration with external healthcare information systems.
12. Continuous evaluation and monitoring of ML model performance after deployment.

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8. Scikit-learn — Official Documentation

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Lundberg, S. M., & Lee, S.-I. (2017). *A Unified Approach to Interpreting Model Predictions.* Advances in Neural Information Processing Systems.

https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html